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Key Points:

- Observational data reveal that falling-tone chorus originates downstream of the magnetic equator while rising-tone chorus arises upstream
- We present the first simulation of falling-tone chorus in a dipole magnetic field, with waves self-consistently excited by energetic electrons
- The simulation shows that falling-tone chirping is associated with positive resonant currents aligned with the wave magnetic field perturbation

Supporting Information:

Supporting Information may be found in the online version of this article.

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Generation of Falling-Tone Whistler-Mode Chorus Waves in the Earth's Magnetosphere

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Abstract Chorus waves are intense whistler-mode emissions in space that play a key role in regulating energetic electron dynamics in the hazardous radiation belt. While rising-tone chorus has been extensively studied, the generation mechanism of falling-tone chorus remains a long-standing problem. Using Van Allen Probe A observations, we reveal that falling-tone chorus in Earth's magnetosphere originates on the downstream side of the magnetic equator along the wave propagation direction, in contrast to the upstream source of rising-tone chorus. Guided by the typical magnetospheric plasma parameters, we perform the first simulation of falling-tone chorus in a dipole magnetic field representative of Earth's magnetosphere and reproduce its downstream source region. We further demonstrate that falling-tone chorus is excited through nonlinear interactions with resonant electrons, where both wave growth and frequency chirping are triggered by resonant currents. Our study fills a critical gap in understanding the nonlinear generation of chorus waves under realistic magnetospheric conditions.

Plain Language Summary Whistler-mode chorus waves are intense electromagnetic emissions widely observed in space and play a significant role in both the formation and depletion of the hazardous radiation belt. A characteristic feature of chorus is its rapid frequency chirping, yet the underlying physical mechanism remains incompletely understood. This is partly because falling-tone chorus, characterized by a decreasing frequency over time, has been less extensively investigated. Using observations from Van Allen Probe A, we find that while both types of chorus are generated near the magnetic equator, falling-tone chorus originates on the downstream side along the wave propagation direction, in contrast to the upstream source of rising-tone chorus. Guided by realistic plasma parameters in observations, we perform a simulation that successfully reproduces falling-tone chorus in the downstream region. The source region is controlled by the balance between the frequency chirping and the background magnetic field gradient, and the wave excitation is determined by nonlinear wave-particle interactions. This work provides the first simulation of falling-tone chorus in a dipole magnetic field and fills a critical gap in understanding its nonlinear generation.

1. Introduction

Chorus waves are intense and coherent whistler-mode emissions that naturally occur in Earth's magnetosphere (Burtis & Helliwell, 1969; Tsurutani & Smith, 1974) and in the magnetospheres of other planets (Coroniti et al., 1980; Harada et al., 2016; Hospodarsky et al., 2008). They are named for their resemblance to a dawn chorus of birds when converted into audible sound. In the Earth's magnetosphere, chorus waves play a crucial role in magnetosphere-ionosphere coupling by driving the precipitation of tens-of-keV electrons from the magnetosphere into the ionosphere (Jaynes et al., 2015; Miyoshi et al., 2015; Kasahara et al., 2018; R. Chen et al., 2024; H. Chen, Wang, Chen, et al., 2025; H. Chen, Wang, Li, et al., 2025), and they also determine the outer radiation belt dynamics by accelerating hundreds-of-keV electrons to relativistic energies (Li et al., 2014; Omura et al., 2015; Summers et al., 1998; Thorne et al., 2013).

A characteristic feature of the chorus spectrum is frequency chirping, in which the central frequency changes rapidly with time (Li et al., 2011; Gao et al., 2014; Santolík et al., 2014; Taubenschuss et al., 2016; Xie et al., 2021; H. Chen et al., 2023; He et al., 2023). Frequency chirping is scientifically important because it helps maintain phase coherence between waves and resonant particles, thereby enabling more efficient particle scattering and acceleration (Omura et al., 2008; Summers et al., 2014; Tsurutani et al., 2020; H. Chen et al., 2024). According to the sign of the frequency chirping rate, chorus waves are classified into rising-tone and falling-tone

chorus (Li et al., 2011; Taubenschuss et al., 2016). Owing to the higher occurrence rate, rising-tone chorus has been extensively investigated through both observations (H. Chen et al., 2023; R. Chen, Miyoshi, et al., 2025; Liu et al., 2025) and simulations (Nunn et al., 1997; Trakhtengerts et al., 2007; Lu et al., 2019; Hanzelka et al., 2020; Ke et al., 2021; Tao et al., 2021; Nogi & Omura, 2023; H. Chen et al., 2024; Wang et al., 2024; Xia et al., 2025), which have demonstrated that its excitation is due to nonlinear physics.

In contrast, only limited simulation attempts have been made to investigate the generation of falling-tone chorus (Fujiwara et al., 2023; Nogi et al., 2020; Nunn & Omura, 2012; Soto-Chavez et al., 2014; Wu et al., 2020, 2022). Moreover, none of these attempts have been performed in a dipole magnetic field configuration representative of Earth's magnetosphere. Specifically, Wu et al. (2020, 2022) simulated falling-tone chorus in a "reversed dipole field," where the magnetic field strength decreases with increasing latitude, opposite to the realistic magnetospheric configuration. This approach originates from Nunn and Omura (2012), who proposed that falling-tone chorus could be generated during wave propagation from higher latitudes toward the magnetic equator. Another approach considers falling-tone chorus excitation in homogeneous magnetic field configurations, where it coexists with rising-tone chorus (Fujiwara et al., 2023; Soto-Chavez et al., 2014; Wu et al., 2020), a feature that is rarely reported in observations. Therefore, simulations of falling-tone chorus generation in a realistic dipole magnetic field remain an important gap in current studies.

In this study, we identify distinct source regions of rising-tone and falling-tone chorus in Earth's magnetosphere using observations from Van Allen Probe A. We perform a self-consistent particle-in-cell (PIC) simulation in a full-scale model of Earth's magnetosphere and successfully reproduce falling-tone chorus waves. The wave source is located downstream of the magnetic equator, consistent with observations. We further quantify the resonant currents contributed by resonant electrons and demonstrate that falling-tone chorus excitation is driven by nonlinear wave-particle interactions. To our knowledge, this work represents the first simulation of falling-tone chorus carried out in a dipole magnetic field.

2. Satellite Observations

2.1. Instrumentations

The Van Allen Probes consist of two identical spacecraft operating at magnetic latitudes within $\sim 20^\circ$ and at L shells extending to $L \sim 7$ (Kessel et al., 2012). The Electric and Magnetic Field Instrument Suite and Integrated Science (EMFISIS; Kletzing et al., 2013) provides waveform measurements at a sampling rate of 35 kHz, with each data segment lasting ~ 6 s. The waveform data of both magnetic and electric fields are analyzed using fast Fourier transforms with a sliding window of 0.03 s and a step size of 0.015 s. EMFISIS also measures the background magnetic field B_0 at a sampling rate of 64 Hz. The plasma density n_{e0} is derived from the upper hybrid frequency measured by the high frequency receiver (Kurth et al., 2015). Spacecraft positions, including the L shell, magnetic local time, and magnetic latitude, are calculated using a dipole magnetic field model.

2.2. Observational Results

Both rising-tone and falling-tone chorus waves are typically observed near $L \sim 6$ (Li et al., 2011; Taubenschuss et al., 2016). Here, we present an example of falling-tone chorus in a 6-s waveform segment detected on 2013-03-01 in the northern hemisphere at $L = 5.83$. Figures 1a–1d show the spectrum of the perturbed magnetic field δB^2 , the ellipticity ε , the wave normal angle θ , and the ratio $P_{\parallel}/|P|$, respectively. Here, $P = (\delta E \times \delta B)/\mu_0$ is the Poynting flux, where δE denotes the perturbed electric field and μ_0 is the vacuum permeability. $P_{\parallel}$ represents the component of P parallel to the background magnetic field. Values of $P_{\parallel}/|P|$ close to 1 (or -1) indicate wave packets propagating along (or against) the background magnetic field.

As shown in Figure 1a, the wave spectrum consists of discrete and repetitive elements with falling-tone chirping. The wave frequency lies below $0.5 f_{ce}$ and decreases from ~ 700 to ~ 350 Hz within ~ 0.20 s. This corresponds to a chirping rate of $\partial f/\partial t = -1.7$ kHz/s, or equivalently $\partial \omega/\partial t = -9.4 \times 10^{-5} \Omega_{e0}^2$. Here, $f_{ce} = eB_{e0}/(2\pi m_e)$ denotes the equatorial electron gyrofrequency, $\Omega_{e0} = 2\pi f_{ce}$ is its angular frequency, B_{e0} is the equatorial background magnetic field strength, e is the elementary charge, and m_e is the electron mass. The mean value of the peak amplitudes is $\sim 10^{-3} B_{e0}$. The waves are right-hand polarized with $\varepsilon \sim 0.8$ (Figure 1b), characteristic of whistler-mode waves. The wave vector propagates slightly obliquely with respect to the background magnetic field, with $\theta < 30^\circ$ (Figure 1c), while the Poynting flux is nearly field-aligned, with $P_{\parallel}/|P| \sim 1$ (Figure 1d).

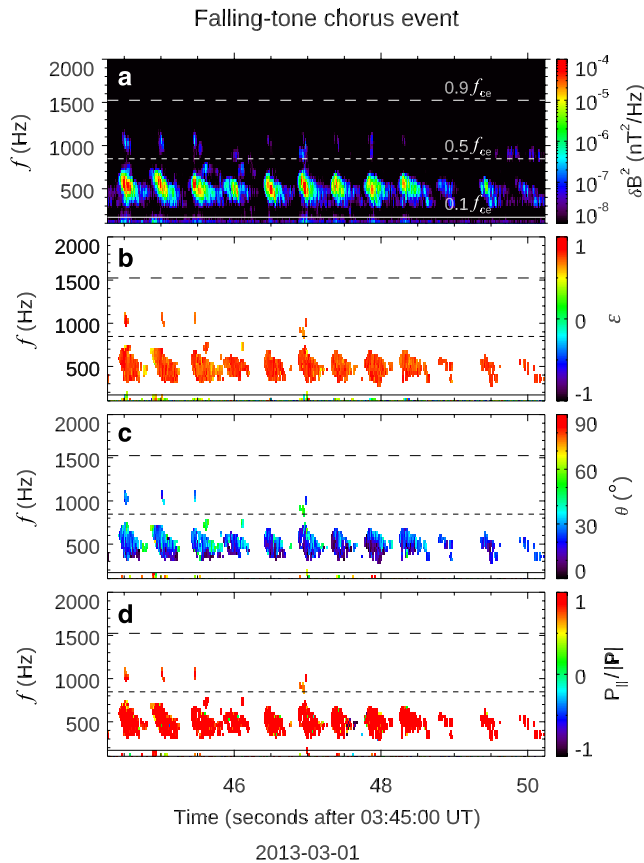


Figure 1. An example of slightly oblique falling-tone chorus waves observed by Van Allen Probe A on 2013-03-01 from 03:45:44.3 to 03:45:50.2 UT at $L = 5.83$, magnetic local time 1.33, and magnetic latitude 5.16° . (a) Spectrum of δB^2 . (b) Ellipticity ϵ . (c) Wave normal angle θ . (d) Ratio $P_{\parallel}/|P|$. Only values of ϵ , θ , and $P_{\parallel}/|P|$ corresponding to $\delta B^2 > 10^{-7}$ nT²/Hz are shown to ensure reliability. Dashed, dotted, and solid lines denote $0.9 f_{ce}$, $0.5 f_{ce}$, and $0.1 f_{ce}$, respectively.

The ratio of the plasma frequency to the equatorial electron gyrofrequency, ω_{pe}/Ω_{ce} , is a key parameter characterizing the plasma condition, where $\omega_{pe} = \sqrt{n_{e0}e^2/m_e\epsilon_0}$, and ϵ_0 is the vacuum permittivity. For this event, ω_{pe}/Ω_{ce} is 10, calculated using the measured $n_{e0} = 3.5$ cm⁻³ and $B_{e0} = 60.5$ nT inferred from the dipole model; the locally measured magnetic field strength is $B_0 = 62.7$ nT. Another slightly oblique falling-tone chorus event with a similar value of $\omega_{pe}/\Omega_{ce} = 8.4$ is presented in Figure S1 in Supporting Information S1.

The repetition of the wave elements is associated with energetic electron injections during substorms (Gao et al., 2022; Lu et al., 2021; Ma et al., 2025), and the weak emissions near 1,000 Hz correspond to the second harmonics waves (Gao et al., 2016; H. Chen et al., 2017). These features are beyond the scope of the present study and are not further examined.

To distinguish the generation of falling-tone chorus from that of rising-tone chorus, we perform a statistical analysis of Van Allen Probe A observations at $L = 3-7$ from October 2012 to December 2014. Chorus waves are identified within the frequency range of $0.1-0.9 f_{ce}$ and are required to have ellipticity greater than 0.7, polarization ratio greater than 0.8, and amplitudes exceeding 5 pT. Each 6-s data segment is treated as an individual chorus event, with rising-tone and falling-tone events classified separately. Events exhibiting both types of chirping are excluded. Using these criteria, a total of 15,941 rising-tone chorus events and 3,755 falling-tone chorus events are identified.

Figures 2a and 2c present the occurrence rate distributions of rising-tone and falling-tone chorus as functions of magnetic latitude λ and ratio $P_{\parallel}/|P|$. For each event, $P_{\parallel}/|P|$ is calculated using a two-step weighted averaging procedure. At each time, $P_{\parallel}/|P|$ is first averaged over frequency using a power weighting, and the wave amplitude is also determined. The $P_{\parallel}/|P|$ for the event is then obtained by applying an amplitude-weighted average over the 6-s interval. The occurrence rate is defined as the number of events in each λ - $P_{\parallel}/|P|$ bin normalized by the total number of events with the same chirping sign. Both rising-tone and falling-tone chorus exhibit $|P_{\parallel}/|P| > 0.7$ for most events, indicating that their Poynting fluxes are nearly field-aligned or anti-

field-aligned. Nevertheless, the two types of waves show notably different latitude distributions relative to the magnetic equator, implying distinct source regions.

For rising-tone chorus propagating against B_0 , although most events are detected in the southern hemisphere, an appreciable occurrence rate extends to latitudes up to $\lambda \sim 3^\circ$ in the northern hemisphere, corresponding to the upstream side of the magnetic equator relative to the wave propagation direction. Similarly, for waves propagating along B_0 , an appreciable occurrence rate extends down to $\lambda \sim -2^\circ$ in the southern hemisphere. These upstream features are more clearly illustrated in the histogram shown in Figure 2b, which presents the distribution of events with $|P_{\parallel}/|P| > 0.7$ as a function of λ . The histogram also indicates that rising-tone chorus exhibits a considerable occurrence rate near the magnetic equator, peaking within $|\lambda| < \sim 5^\circ$.

In contrast, falling-tone chorus exhibits much lower occurrence rates near the magnetic equator. Almost all falling-tone events are detected on the downstream side of the magnetic equator, with events propagating against B_0 occurring predominantly in the southern hemisphere and those propagating along B_0 mainly in the northern hemisphere. As shown in Figure 2d, the peak occurrence rate is located at $|\lambda| \sim 6^\circ-10^\circ$, well away from the magnetic equator. This comparison indicates that rising-tone chorus originates on the upstream side of the magnetic equator, whereas falling-tone chorus occurs on the downstream side. These results obtained in the inner magnetosphere with $L < \sim 6$ are consistent with observations by Time History of Events and Macroscale Interactions during Substorms (THEMIS) in the outer magnetosphere (Taubenschuss et al., 2016).

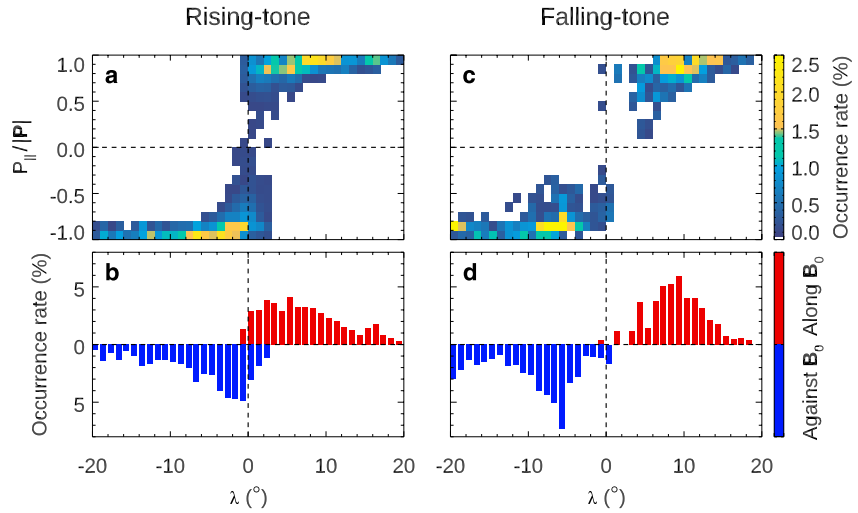


Figure 2. Statistical results of rising-tone (left column) and falling-tone chorus waves (right column) observed by Van Allen Probe A from October 2012 to December 2014. (a, c) Occurrence rates as functions of λ and $P_{||}/|P|$. (b, d) Event occurrence versus λ . Red (blue) denotes events propagating along (against) the background magnetic field with $P_{||}/|P| > 0.7$ (< -0.7). Horizontal dotted lines indicate $P_{||}/|P| = 0$ in (a, c) and zero occurrence in (b, d), and vertical dotted lines in all panels mark the magnetic equator.

The upstream source region of rising-tone chorus is widely attributed to nonlinear wave-particle interactions (Demekhov et al., 2020; Hanzelka et al., 2020; Tao et al., 2021; Nogi & Omura, 2023; H. Chen et al., 2023; Wang et al., 2024), whereas the generation of falling-tone chorus remains poorly understood. Using the plasma condition $\omega_{pe}/\Omega_{ce} = 10$ derived from the falling-tone chorus event shown in Figure 1, we perform a PIC simulation to examine the underlying physics responsible for falling-tone chorus excitation.

3. PIC Simulation in a Dipole Magnetic Field

3.1. Simulation Model

The simulation is conducted in the two-dimensional (2-D) meridian plane using the general curvilinear plasma simulation code (GCPIC) model (Lu et al., 2019; Wang et al., 2024). The model adopts a background dipole magnetic field and solves Maxwell's equations together with particle dynamics in dipole coordinates. Particle trajectories are advanced by following their three-dimensional velocities and positions under the combined influence of the background magnetic field and the perturbed electromagnetic fields. Background electrons are treated as a fluid, whereas energetic electrons are tracked as kinetic particles under the relativistic Lorentz force. Ions are immobile. Reflecting boundary conditions are imposed on particles, and absorbing boundary conditions are applied to waves (H. Chen, Wang, Li, et al., 2025).

The simulation is performed in a full-scale dipole magnetic field, with the domain spanning a radial range of $L = 5.95$ – 6.05 and a latitudinal range of $\lambda = \pm 20^\circ$. The model adopts typical magnetospheric parameters and incorporates the plasma condition $\omega_{pe}/\Omega_{e0} = 10$ from the observational event. The background electron number density n_{e0} is uniform with $n_{e0} = 20 \text{ cm}^{-3}$ (Denton et al., 2004), and the background magnetic field strength at the domain center $L_0 = 6$ is $B_{e0} = 144.4 \text{ nT}$. At this location, the ratio $\omega_{pe}/\Omega_{e0} = 10$, identical to that in the falling-tone chorus event shown in Figure 1. Energetic electrons are prescribed with a loss-cone distribution, implemented using a subtracted bi-Maxwellian distribution (Zenitani & Nakano, 2023):

$$f(u_{||}, u_{\perp}) = \frac{n_{heq}}{(2\pi)^{3/2} U_{||} U_{\perp}^2 (1 - \rho\beta)} \exp\left(-\frac{u_{||}^2}{2U_{||}^2}\right) \cdot \left[\exp\left(-\frac{u_{\perp}^2}{2U_{\perp}^2}\right) - \rho \exp\left(-\frac{u_{\perp}^2}{2\beta U_{\perp}^2}\right) \right], \quad (1)$$

where $\rho = 1$ and $\beta = 0.3$ are constant parameters, with n_{heq} denoting the equatorial energetic-electron density, $U_{||}$ and $U_{\perp}$ the parallel and perpendicular thermal momentum, and $u_{||}$ and $u_{\perp}$ the parallel and perpendicular

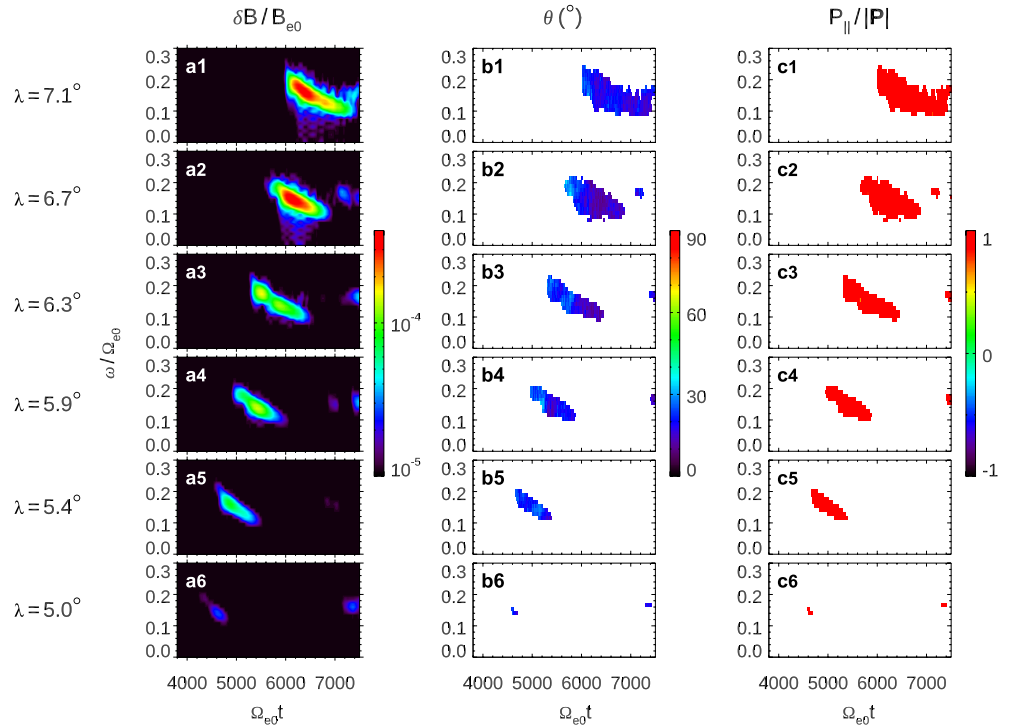


Figure 3. Falling-tone chorus waves in the simulation. (a) Wave spectrum of δB . (b) Wave normal angle θ . (c) Ratio $P_{||}/|P|$. The panels are ordered from top to bottom by magnetic latitude: $\lambda = 7.1^\circ$, 6.7° , 6.3° , 5.9° , 5.4° and 5.0° , corresponding to the downstream side of the magnetic equator. In panels (b) and (c), only values of θ and $P_{||}/|P|$ corresponding to $\delta B/B_{e0} > 4 \times 10^{-5}$ are shown.

momentum. The n_{heq} is set as $0.005 n_{e0}$ (Gao et al., 2014; Rasinskaite et al., 2025). The parallel temperature is $m_e U_{||}^2 = 20 \text{ keV}$ (Rasinskaite et al., 2025), and the temperature anisotropy is $U_{\perp}^2/U_{||}^2 = 1.44$ (Fu et al., 2014; Yue et al., 2016). The latitudinal distribution of energetic electrons follows Summers et al. (2012). The background electrons satisfy a Maxwellian distribution.

The simulation domain consists of $400 \times 57,600$ grids in the curvilinear p - q plane, where $\hat{p}$ and $\hat{q}$ correspond to the radial and latitudinal directions, respectively. The average grid spacings are $\Delta p = 1.34 V_{Ae0}/\Omega_{e0}$ and $\Delta q = 0.34 V_{Ae0}/\Omega_{e0}$. Here, $V_{Ae0} = B_{e0}/\sqrt{\mu_0 n_{e0} m_e}$ denotes the electron Alfvén speed, and μ_0 is the vacuum permeability. The total particle number is 2×10^9 for each species, and the time step is $\Delta t = 0.025 \Omega_{e0}^{-1}$. The δf method is used to reduce numerical noise (Lu et al., 2021).

3.2. Simulation Results

Figures 3a1–3a6 illustrate the spatial evolution of the wave spectrum δB at $L = 6.03$ in the northern hemisphere, where falling-tone chirping is present across all latitudes. The falling-tone signature is also evident in the instantaneous frequency shown in Figure 4b. The waves propagate slightly obliquely with $\theta < 30^\circ$ (Figures 3b1–3b6), while the Poynting flux remains nearly field-aligned with $P_{||}/|P| \sim 1$ (Figures 3c1–3c6). The right-hand polarization is confirmed in Figure S2 in Supporting Information S1. These properties agree with the falling-tone chorus event shown in Figure 1.

Falling-tone chorus is confined to the downstream side of the magnetic equator. It first appears at $\lambda = 5.0^\circ$ with a weak wave amplitude of $\sim 3 \times 10^{-4} B_{e0}$ (Figure 3a6). During propagation toward higher latitudes, nonlinear interactions with resonant electrons lead to wave growth, with the wave amplitude increasing to $\sim 2 \times 10^{-3} B_{e0}$ at $\lambda = 7.1^\circ$, comparable to the values reported in observations (Figure 1). A similar evolution occurs in the southern hemisphere, as shown in Figure S3 in Supporting Information S1. The simulated downstream falling-tone chorus is consistent with observational statistics (Figures 2c and 2d).

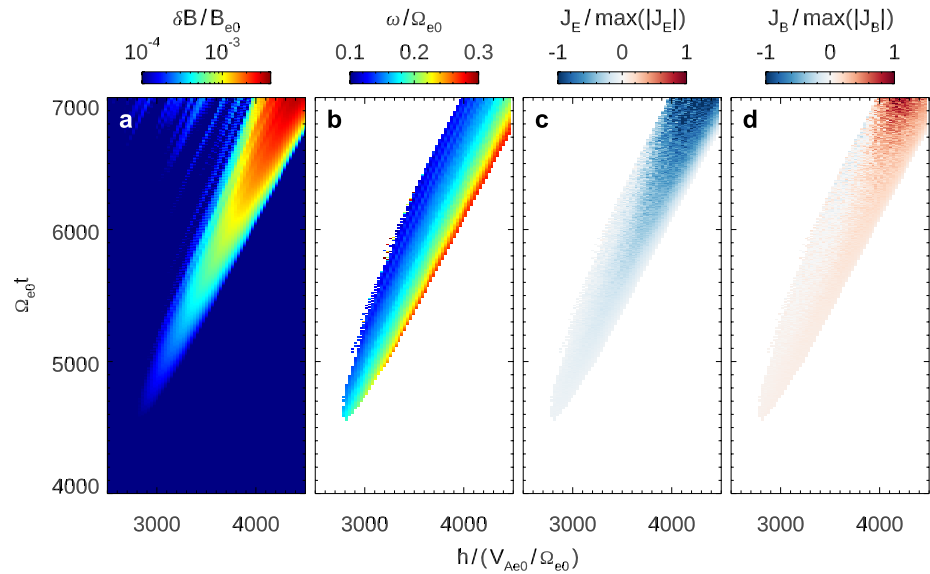


Figure 4. Simulated resonant currents. (a) Wave amplitude $\delta B/B_{e0}$ and (b) instantaneous wave frequency ω/Ω_{e0} . Resonant currents (c) J_E and (d) J_B , each normalized to its respective maximum value. To reduce noise, the values in panels (b)–(d) corresponding to $\delta B/B_{e0} < 2 \times 10^{-4}$ are not shown.

The inhomogeneity factor S is a key parameter in nonlinear theory and characterizes the efficiency of energy exchange between waves and resonant particles (Nunn, 1974; Vomvoridis et al., 1982; Zonca et al., 2022). It is defined as (Omura, 2021):

$$S = -\frac{m_e}{s_0 e \omega \delta B} \left(s_1 \frac{\partial \omega}{\partial t} + c s_2 \frac{\partial B_0}{\partial h} \right), \quad (2)$$

where

$$s_0 = \frac{\xi c}{\chi v_{\perp}}, \quad (3)$$

$$s_1 = \gamma \left(1 - \frac{v_c}{v_g} \right)^2, \quad (4)$$

$$s_2 = \frac{1}{2\xi\chi} \left\{ \frac{\gamma\omega}{\Omega_e} \left(\frac{v_{\perp}}{c} \right)^2 - \left[2 + \frac{\omega}{\Omega_e} \frac{\chi^2(\Omega_e - \gamma\omega)}{\Omega_e - \omega} \right] \frac{v_c v_p}{c^2} \right\}. \quad (5)$$

Here, $\xi = \sqrt{\omega(\Omega_e - \omega)}/\omega_{pe}$ and $\chi = \sqrt{1 - \omega^2/(c^2 k^2)}$, c is the speed of light, Ω_e is the local electron gyro-frequency, γ is the relativistic factor, $v_c = (\omega - \Omega_e/\gamma)/(k \cos \theta)$ is the cyclotron resonant velocity, v_p and v_g are the phase and group velocities, and $v_{\perp}$ is the perpendicular electron velocity. The coefficients s_0 , s_1 , and s_2 are all positive.

The inhomogeneity factor S consists of two contributions, namely a frequency-chirping term proportional to $\partial\omega/\partial t$ and a background magnetic-field gradient term proportional to $\partial B_0/\partial h$, where h is the distance along the field line to the magnetic equator. We estimate S at $\lambda = 7.1^\circ$, where the peak wave amplitude reaches $1.8 \times 10^{-3} B_{e0}$, corresponding to a frequency of $0.16 \Omega_{e0}$ (see Figure S3 in Supporting Information S1) and a wave normal angle of 15.2° . The chirping rate is $\partial\omega/\partial t = -9.6 \times 10^{-5} \Omega_{e0}^2$, agreeing well with the observational event. The background magnetic field gradient at this position is $\partial B_0/\partial h = 3.7 \times 10^{-5} B_{e0}/(V_{Ae0}/\Omega_{e0})$. Substituting these parameters yields $S = 0.54$, which is consistent with the theoretical expectation of $S \sim 0.4$, a value that

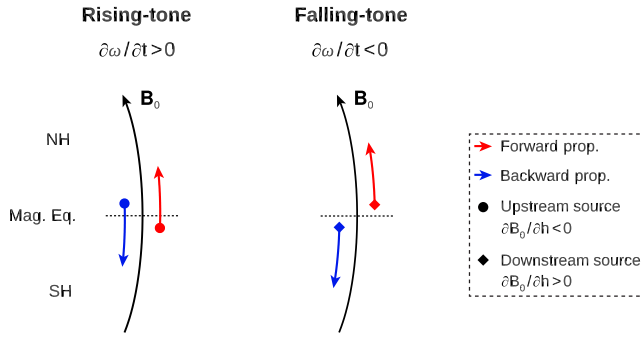


Figure 5. Schematic of chorus source regions. The left column illustrates rising-tone chorus with positive $\partial\omega/\partial t$, whereas the right column illustrates falling-tone chorus with negative $\partial\omega/\partial t$. The background magnetic field B_0 is shown by the black curves with arrows, and the magnetic equator (Mag. Eq.) is indicated by the horizontal dashed lines. NH and SH denote the northern and southern hemispheres, respectively. Red arrows represent forward propagating waves, and blue arrows represent backward propagating waves. Circles indicate the upstream source region for rising-tone chorus, where $\partial B_0/\partial h < 0$, and diamonds indicate the downstream source region for falling-tone chorus, where $\partial B_0/\partial h > 0$.

corresponds to the most efficient energy transfer between chorus waves and resonant electrons (Omura et al., 2008).

Resonant electrons trapped in the wave gyrophases contribute to resonant currents, which are responsible for the nonlinear excitation of chorus waves (Omura, 2021; Zonca et al., 2022). According to nonlinear theory, the resonant current J_E , directed along the perturbed electric field $\delta E_\perp$, is suggested to characterize the wave-particle energy transfer, with negative values corresponding to wave amplification. The resonant current J_B , directed along the perturbed magnetic field $\delta B_\perp$, is proposed to determine the chirping direction, such that positive J_B leads to a falling-tone chirping (see Equation 74 in Omura (2021)). Figures 4c and 4d present the resonant currents J_E and J_B in the h - t planes for energetic electrons, defined as

$$J_E = \int_0^\infty \int_0^{2\pi} \int_{-\infty}^\infty -e v_\perp \cdot \delta \hat{E}_\perp \delta f(u_\parallel, \zeta, u_\perp) u_\perp du_\parallel d\zeta du_\perp, \quad (6)$$

$$J_B = \int_0^\infty \int_0^{2\pi} \int_{-\infty}^\infty -e v_\perp \cdot \delta \hat{B}_\perp \delta f(u_\parallel, \zeta, u_\perp) u_\perp du_\parallel d\zeta du_\perp, \quad (7)$$

where $\delta \hat{E}_\perp$ and $\delta \hat{B}_\perp$ denote the unit vectors of the perturbed electromagnetic fields. Figures 4a and 4b also present the wave amplitude $\delta B/B_{e0}$ and the instantaneous frequency ω/Ω_{e0} for reference. The instantaneous frequency is obtained from adjacent zero-crossing of the azimuthal magnetic field perturbation δB_ϕ , following the method of Tsurutani et al. (2020).

As shown in Figure 4c, J_E remains negative, indicating that energetic electrons transfer energy to waves, leading to wave growth along the propagation path (Figure 4a). Meanwhile, positive J_B is established, which leads to a gradual decrease in wave frequency with time (Omura, 2021; Zonca et al., 2022), producing falling-tone chirping (Figure 4b). The positive J_B is consistent with a positive inhomogeneity factor S , under which resonant electrons are trapped within the gyrophase range $\zeta \sim \pi-3\pi/2$ (Omura et al., 2015), where ζ denotes the phase angle between $v_\perp$ of resonant electrons and the wave magnetic field perturbation $\delta B_\perp$. In this range, $v_\perp \cdot \delta \hat{B}_\perp$ is negative, leading to a positive J_B according to Equation 7. These quantifications demonstrate that the generation of falling-tone chorus waves is determined by nonlinear wave-particle interactions.

4. Discussions

Distinguishing between rising-tone and falling-tone chorus is essential for advancing our understanding of nonlinear wave-particle interactions. Figure 5 presents a schematic illustration summarizing the source regions of the two types of chorus waves, in which rising-tone chorus originates upstream of the magnetic equator, whereas falling-tone chorus is generated downstream. The distinct source regions of rising-tone and falling-tone chorus can be interpreted in terms of nonlinear physics through the definition of inhomogeneity factor S . At the source region, while S is nonzero, the wave amplitude is extremely weak, such that the product $S\delta B$ approaches zero. From Equation 2, the frequency-chirping term is balanced by the background magnetic-field inhomogeneity term. For rising-tone chorus, a positive frequency chirping rate $\partial\omega/\partial t$ requires a negative background magnetic-field gradient $\partial B_0/\partial h$ at the source region, corresponding to the upstream side of the magnetic equator. This region is indicated by the circles in Figure 5, where the background magnetic field strength decreases along the wave propagation direction. In contrast, for falling-tone chorus, a negative $\partial\omega/\partial t$ is compensated by a positive $\partial B_0/\partial h$, leading to the source region downstream of the magnetic equator. This region is denoted by the diamonds in Figure 5, where B_0 increases along the wave propagation direction.

The downstream source region of falling-tone chorus obtained in our simulation is consistent with spacecraft observations (Figures 2c and 2d). In contrast, the falling-tone chorus simulated by Wu et al. (2022) originates on the upstream side of the magnetic equator. Nunn and Omura (2012) simulated the generation of falling-tone chorus during propagation from higher latitudes toward the magnetic equator, whereas observations indicate that chorus waves generally propagate toward higher latitudes.

The initial setup adopted in the present simulation differs from those used in previous 2-D simulations of rising-tone chorus (Lu et al., 2019; Wang et al., 2024). Both the observational event (Figure 1) and the simulation results indicate that slightly oblique falling-tone chorus occurs at relatively large $\omega_{pe}/\Omega_{e0} \sim 10$, whereas the rising-tone chorus simulations of Lu et al. (2019) and Wang et al. (2024) employed values around 5. Other plasma parameters are broadly comparable. To understand the underlying mechanism, a detailed investigation of the nonlinear particle dynamics associated with falling-tone chorus is required. Future parametric studies will examine the conditions governing the transition from falling-tone to rising-tone chorus.

Another difference is that the present simulation is performed in a full-scale dipole magnetic field, whereas Lu et al. (2019) and Wang et al. (2024) adopted a reduced-scale configuration. The full-scale configuration allows quantitative comparisons with observations and reproduces the observed wave normal angle, frequency chirping rate, and wave amplitude.

Based on plasma parameters constrained by a slightly oblique falling-tone chorus event, we reproduce falling-tone chorus in the simulation. Previous observations indicate that falling-tone chorus is more frequently associated with highly oblique propagation (Li et al., 2011). The statistical distribution of wave normal angles for falling-tone chorus events is provided in Figure S4 in Supporting Information S1. Simulations of highly oblique falling-tone chorus events will be pursued in future work.

In the present study, a dipole magnetic field is adopted as the background field because it represents the typical magnetic field configuration of Earth's magnetosphere. However, existing nonlinear theories of chorus excitation are not restricted to dipole magnetic fields and primarily require magnetic field inhomogeneity along the field line (Nunn, 1974; Omura et al., 2008; Tao et al., 2021; Vomvoridis et al., 1982; Zonca et al., 2022). Therefore, the nonlinear physics associated with chorus source region may also be applicable to non-dipolar inhomogeneous magnetic-field configurations under appropriate plasma parameters, such as in the Earth's magnetotail, as well as to other magnetized planets with intrinsic magnetic fields, including Saturn and Jupiter, where rising-tone chorus emissions have been reported (Hospodarsky et al., 2008; Liu et al., 2025; Menietti et al., 2016). Future observations are needed to determine whether falling-tone chorus also occurs in these environments and, if so, how its properties differ from those of rising-tone chorus.

5. Summary

Using Van Allen Probe A observations from October 2012 to December 2014, we reveal distinct source regions for rising-tone and falling-tone chorus in Earth's magnetosphere. While both types are generated near the magnetic equator, falling-tone chorus originates on the downstream side along the wave propagation direction, in contrast to the upstream source of rising-tone chorus.

Based on typical magnetospheric plasma parameters, we perform the first simulation of falling-tone chorus in a dipole magnetic field. The waves are self-consistently excited by energetic electrons and are confined to the downstream side of the magnetic equator, consistent with observations. The simulation further indicates that the downstream source region of falling-tone chorus arises from nonlinear wave-particle interactions, with wave growth associated with negative resonant currents J_E and falling-tone chirping associated with positive resonant currents J_B .

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All Van Allen Probe A data are available from <https://spdf.gsfc.nasa.gov/pub/data/rbsp/rbspa/>. The observation data were analyzed using the Space Physics Environment Data Analysis System (SPEDAS; Angelopoulos et al., 2019). The simulation data used in this study are available in H. Chen and Wang (2026).

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